

Evaluation of Depression Caused by Haloperidol Which is Used to Induce Catalepsy in Albino Mice

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Master of Pharmacy

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CERTIFICATE

This is to certify that the thesis entitled "Evaluation of Depression Caused by Haloperidol Which is Used to Induce Catalepsy in Albino Mice" submitted by [Student Name] to [Institution Name] for the award of the degree of Master of Pharmacy in Pharmacology is a bonafide work carried out by the candidate under my direct supervision and guidance. This work has not been submitted earlier for any degree or diploma to any other University or Institution.

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DECLARATION

I hereby declare that the thesis entitled "Evaluation of Depression Caused by Haloperidol Which is Used to Induce Catalepsy in Albino Mice" submitted to [Institution Name] for the award of the degree of Master of Pharmacy in Pharmacology is a record of original and genuine research work carried out by me under the supervision of [Guide Name]. I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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TABLE OF CONTENTS

1. INTRODUCTION	8
2. LITERATURE REVIEW	12
3. AIM AND OBJECTIVES	18
4. PLAN OF WORK	19
5. MATERIALS AND METHODS	20
6. RESULTS	26
7. DISCUSSION	32
8. CONCLUSION	36
9. BIBLIOGRAPHY	38
10. APPENDICES	42

LIST OF ABBREVIATIONS

- CNS: Central Nervous System
- DA: Dopamine
- EPS: Extrapyramidal Symptoms
- FST: Forced Swimming Test
- GABA: Gamma-Aminobutyric Acid
- HAL: Haloperidol

- 5-HT: 5-Hydroxytryptamine (Serotonin)
 - IAEC: Institutional Animal Ethics Committee
 - i.p.: Intraperitoneal
 - NE: Norepinephrine
 - OECD: Organisation for Economic Co-operation and Development
 - PD: Parkinson's Disease
 - TST: Tail Suspension Test
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1. INTRODUCTION

Depression represents one of the most prevalent and debilitating psychiatric disorders worldwide, affecting millions of individuals across diverse demographic populations. The World Health Organization has consistently ranked depression among the leading causes of disability globally, with its impact extending far beyond individual suffering to encompass significant societal and economic burdens. The complex etiology of depression involves intricate interactions between genetic predisposition, environmental factors, neurochemical imbalances, and psychosocial stressors, making it a multifaceted condition that requires comprehensive understanding for effective therapeutic intervention.

The neurobiological foundations of depression have been extensively studied, revealing disruptions in multiple neurotransmitter systems, including serotonin, norepinephrine, and dopamine pathways. Among these, the dopaminergic system has emerged as a crucial player in the pathophysiology of depression, particularly in the context of anhedonia, motivation deficits, and psychomotor retardation. The dopaminergic circuits, encompassing the mesocorticolimbic and nigrostriatal pathways, are intimately involved in reward processing, motor function, and emotional regulation, making them prime targets for both therapeutic intervention and potential iatrogenic effects.

Haloperidol, a prototypical first-generation antipsychotic medication belonging to the butyrophenone class, has been widely utilized in clinical practice for the management of schizophrenia, acute psychotic episodes, and various movement disorders. Its primary mechanism of action involves potent antagonism of dopamine D2 receptors, particularly in the nigrostriatal and mesolimbic pathways. While this dopaminergic blockade is therapeutically beneficial for controlling positive symptoms of psychosis, it simultaneously predisposes patients to a spectrum of motor and non-motor adverse effects, including extrapyramidal symptoms, tardive dyskinesia, and notably, depressive symptomatology.

The relationship between dopamine receptor antagonism and the development of depressive symptoms has been a subject of intense scientific inquiry. Clinical observations have consistently documented the occurrence of depression in patients receiving antipsychotic medications, with haloperidol being particularly implicated due to its high potency and selectivity for dopamine D2 receptors. The incidence of antipsychotic-induced depression varies considerably across studies, ranging from 20% to 60% of

treated patients, depending on the specific medication, dosage regimen, treatment duration, and individual patient factors.

Catalepsy, characterized by the maintenance of imposed postures for extended periods, represents a well-established animal model for studying extrapyramidal side effects associated with antipsychotic medications. This phenomenon occurs due to the disruption of dopaminergic neurotransmission in the nigrostriatal pathway, leading to impaired motor function and postural rigidity. The catalepsy model has been extensively validated as a predictor of extrapyramidal liability for antipsychotic compounds and serves as an essential tool in preclinical drug development and mechanistic studies.

The utilization of haloperidol to induce catalepsy in laboratory animals has become a standard experimental paradigm, providing researchers with a reproducible and quantifiable method to investigate the motor effects of dopamine receptor antagonism. However, the broader neurobiological consequences of haloperidol administration, particularly its potential to induce depressive-like behaviors, have received relatively less attention in preclinical research. This gap in knowledge represents a significant limitation in our understanding of the comprehensive effects of antipsychotic medications and their impact on mood regulation.

Albino mice, as experimental subjects, offer several advantages for neuropharmacological research, including genetic homogeneity, cost-effectiveness, ease of handling, and well-characterized behavioral repertoires. The availability of validated behavioral assays for assessing depressive-like behaviors in mice, such as the forced swimming test, tail suspension test, and sucrose preference test, facilitates the systematic evaluation of mood alterations induced by pharmacological interventions.

The present study addresses a critical gap in the current literature by systematically investigating the depressive effects of haloperidol administration in the context of catalepsy induction. While previous research has primarily focused on the motor manifestations of haloperidol treatment, the concurrent evaluation of mood-related behavioral changes remains underexplored. This investigation is particularly relevant given the clinical observations of depression in patients receiving haloperidol therapy and the need for comprehensive preclinical models that capture both motor and mood-related adverse effects.

The experimental approach employed in this study involves the administration of haloperidol to albino mice at doses sufficient to induce catalepsy, followed by comprehensive behavioral assessment using validated paradigms for evaluating depressive-like behaviors. This dual approach allows for the simultaneous characterization of motor and mood-related effects, providing insights into the mechanistic relationship between dopaminergic dysfunction and depression.

Understanding the depressive effects of haloperidol in the context of catalepsy induction has significant implications for clinical practice, drug development, and our fundamental understanding of the neurobiological basis of depression. From a clinical perspective, this knowledge can inform healthcare providers about the potential mood-related side effects of haloperidol therapy, enabling more informed treatment decisions and improved patient monitoring strategies. In the realm of drug development, these

findings can contribute to the design of novel antipsychotic agents with reduced propensity for inducing depressive symptoms.

Furthermore, this research contributes to the broader scientific understanding of the relationship between dopaminergic dysfunction and depression. The dopamine system's role in depression has gained increasing recognition, with emerging evidence suggesting that dopaminergic deficits may be particularly relevant to specific symptom clusters, including anhedonia, psychomotor retardation, and motivational deficits. By systematically examining the depressive consequences of controlled dopaminergic blockade in an animal model, this study provides valuable mechanistic insights into the dopamine-depression relationship.

The methodology employed in this investigation adheres to internationally recognized standards for animal research, ensuring the ethical conduct of experiments while maximizing the scientific validity of the findings. The use of multiple behavioral assays enhances the reliability and comprehensiveness of the depression assessment, while careful attention to experimental design and statistical analysis ensures the robustness of the conclusions.

In conclusion, this thesis presents a comprehensive investigation into the evaluation of depression caused by haloperidol in the context of catalepsy induction in albino mice. The study addresses a significant gap in the current literature and provides valuable insights into the mood-related effects of dopamine receptor antagonism. The findings contribute to our understanding of antipsychotic-induced depression and have implications for clinical practice, drug development, and basic neuroscience research.

2. LITERATURE REVIEW

2.1 Historical Perspective on Haloperidol and Antipsychotic Medications

The discovery and development of haloperidol represents a pivotal milestone in the evolution of psychopharmacology and the treatment of psychiatric disorders. Haloperidol was first synthesized by Paul Janssen in 1958 at Janssen Pharmaceutica in Belgium, marking the beginning of a new era in antipsychotic therapy. The compound was developed as part of a systematic medicinal chemistry program aimed at creating more potent and selective antipsychotic agents compared to the existing phenothiazines such as chlorpromazine.

The introduction of haloperidol into clinical practice in the early 1960s revolutionized the treatment of schizophrenia and other psychotic disorders. Its high potency, with therapeutic efficacy achieved at relatively low doses compared to low-potency antipsychotics, made it an attractive option for clinicians. The drug's long half-life and availability in both oral and injectable formulations, including a long-acting depot preparation, further enhanced its clinical utility and contributed to improved medication adherence in patients with chronic psychiatric conditions.

Throughout the subsequent decades, haloperidol became one of the most widely prescribed antipsychotic medications globally, with extensive clinical experience accumulated across diverse patient

populations and psychiatric conditions. Its efficacy in managing acute psychotic episodes, controlling agitation, and preventing relapse in schizophrenia has been well-documented through numerous controlled clinical trials and real-world studies. However, this widespread clinical use also revealed the significant burden of adverse effects associated with potent dopamine receptor antagonism.

2.2 Pharmacological Profile of Haloperidol

Haloperidol belongs to the butyrophenone class of antipsychotic medications and is characterized by its potent and selective antagonism of dopamine D2 receptors. The drug exhibits high affinity for D2 receptors with a dissociation constant (K_i) of approximately 1-3 nM, making it one of the most potent dopamine receptor antagonists available in clinical practice. This high potency translates to therapeutic efficacy at relatively low doses, typically ranging from 2-20 mg per day in adult patients.

The pharmacokinetic profile of haloperidol is characterized by good oral bioavailability (approximately 60-70%), extensive hepatic metabolism via cytochrome P450 enzymes (primarily CYP3A4 and CYP2D6), and a relatively long elimination half-life of 18-54 hours in healthy individuals. The drug undergoes significant first-pass metabolism, with the primary metabolite being reduced haloperidol, which retains some pharmacological activity but is generally considered less potent than the parent compound.

Haloperidol's mechanism of action extends beyond simple D2 receptor antagonism, as the drug also exhibits affinity for other receptor systems, although to a lesser extent. These include modest antagonism of α_1 -adrenergic receptors, which may contribute to orthostatic hypotension and sedation, and minimal interaction with cholinergic, histaminergic, and serotonergic receptors compared to other antipsychotic agents. This relatively selective pharmacological profile contributes to haloperidol's characteristic side effect profile, which is predominantly related to dopaminergic blockade.

2.3 Dopaminergic System and Its Role in Motor Function and Mood Regulation

The dopaminergic system comprises several anatomically and functionally distinct pathways that play crucial roles in motor control, reward processing, motivation, and emotional regulation. The four major dopaminergic pathways include the nigrostriatal, mesolimbic, mesocortical, and tuberoinfundibular systems, each serving specific physiological functions and exhibiting differential susceptibility to pharmacological manipulation.

The nigrostriatal pathway, originating from dopaminergic neurons in the substantia nigra pars compacta and projecting to the striatum (caudate nucleus and putamen), is primarily responsible for the initiation and control of voluntary movement. This pathway is the primary target for the motor effects of antipsychotic medications, with dopamine receptor blockade leading to extrapyramidal symptoms including catalepsy, dystonia, parkinsonism, and tardive dyskinesia. The delicate balance of dopaminergic neurotransmission in this pathway is essential for normal motor function, and disruption of this balance results in the characteristic movement disorders associated with antipsychotic therapy.

The mesolimbic pathway, extending from the ventral tegmental area to limbic structures including the nucleus accumbens, amygdala, and hippocampus, plays a central role in reward processing, motivation,

and the pathophysiology of addiction. This pathway is also implicated in the positive symptoms of schizophrenia, making it a primary therapeutic target for antipsychotic medications. However, excessive blockade of dopamine receptors in this pathway can lead to anhedonia, motivational deficits, and depressive symptoms, highlighting the fine line between therapeutic efficacy and adverse effects.

The mesocortical pathway, projecting from the ventral tegmental area to various cortical regions including the prefrontal cortex, is involved in cognitive function, working memory, and executive control. Dysfunction of this pathway is associated with the negative symptoms and cognitive deficits observed in schizophrenia. Paradoxically, this pathway often exhibits reduced dopaminergic activity in schizophrenia, and further suppression by antipsychotic medications may exacerbate cognitive impairment and contribute to secondary negative symptoms.

2.4 Catalepsy as an Animal Model of Extrapyramidal Side Effects

Catalepsy represents a well-established and widely utilized animal model for investigating the extrapyramidal side effects associated with antipsychotic medications. This phenomenon is characterized by the animal's inability to correct externally imposed postures, resulting in the maintenance of unusual positions for extended periods. The cataleptic response is considered to reflect the disruption of normal motor function due to dopaminergic blockade in the nigrostriatal pathway.

The catalepsy model has been extensively validated through numerous studies demonstrating a strong correlation between the cataleptic potency of antipsychotic drugs in animals and their propensity to cause extrapyramidal side effects in humans. This predictive validity has made catalepsy testing an essential component of preclinical screening programs for antipsychotic drug development, allowing researchers to identify compounds with reduced extrapyramidal liability before progressing to clinical trials.

Various methods have been developed to assess catalepsy in laboratory animals, with the most commonly used approach involving the placement of the animal's forepaws on an elevated horizontal bar and measuring the duration for which this imposed posture is maintained. The cataleptic response typically develops within 30-60 minutes following antipsychotic administration and can persist for several hours, depending on the drug's potency, dose, and pharmacokinetic properties.

The neurobiological basis of catalepsy involves the disruption of dopaminergic neurotransmission in the dorsal striatum, leading to an imbalance between dopaminergic and cholinergic activity. This imbalance results in altered basal ganglia output and subsequent disruption of motor control circuits. The severity of catalepsy is generally dose-dependent and correlates with the degree of striatal dopamine D2 receptor occupancy, with significant catalepsy typically occurring at receptor occupancy levels exceeding 80%.

2.5 Depression and Its Neurobiological Underpinnings

Depression is a complex psychiatric disorder characterized by persistent feelings of sadness, hopelessness, loss of interest or pleasure in activities, and a range of cognitive, physical, and behavioral symptoms that significantly impair daily functioning. The lifetime prevalence of major depressive disorder

is estimated to be approximately 10-15% in the general population, with women being affected twice as frequently as men.

The neurobiological basis of depression involves dysregulation of multiple neurotransmitter systems, including serotonin, norepinephrine, and dopamine. The traditional monoamine hypothesis of depression proposed that the condition results from deficiencies in brain monoamine neurotransmitters, particularly serotonin and norepinephrine. However, this hypothesis has evolved to incorporate more complex interactions between neurotransmitter systems and additional factors such as neuroplasticity, inflammation, and stress response mechanisms.

The role of dopamine in depression has gained increasing recognition over the past two decades, with accumulating evidence suggesting that dopaminergic dysfunction may be particularly relevant to specific symptom clusters of depression. Anhedonia, the inability to experience pleasure from previously rewarding activities, is considered a core symptom of depression and is closely linked to dopaminergic function in the mesolimbic reward pathway. Similarly, psychomotor retardation, fatigue, and motivational deficits observed in depression may reflect dopaminergic hypofunction in various brain circuits.

Neuroimaging studies in depressed patients have revealed alterations in dopaminergic function, including reduced dopamine synthesis capacity in the striatum, altered dopamine transporter availability, and abnormal activation patterns in dopamine-innervated brain regions during reward-related tasks. These findings support the hypothesis that dopaminergic dysfunction contributes to the pathophysiology of depression and may represent a therapeutic target for antidepressant interventions.

2.6 Animal Models of Depression

Animal models have played a crucial role in advancing our understanding of depression and in the development of antidepressant medications. These models are designed to reproduce specific aspects of human depression, including behavioral, neurochemical, and physiological changes observed in depressed patients. While no single animal model can fully capture the complexity of human depression, well-validated models provide valuable tools for investigating the mechanisms underlying depressive disorders and evaluating potential therapeutic interventions.

The forced swimming test (FST), originally developed by Porsolt and colleagues in 1977, is one of the most widely used behavioral assays for assessing antidepressant activity in rodents. The test is based on the observation that when rodents are placed in an inescapable cylinder filled with water, they initially exhibit vigorous swimming and climbing behaviors in attempts to escape, but eventually develop a characteristic immobility behavior interpreted as behavioral despair or resignation. The duration of immobility is considered a measure of depressive-like behavior, with antidepressant treatments typically reducing immobility time.

The tail suspension test (TST) represents another commonly employed behavioral assay for evaluating depressive-like behaviors in mice. In this test, mice are suspended by their tails from a horizontal bar, and the duration of immobility is measured. Similar to the FST, immobility in the TST is interpreted as a

measure of behavioral despair, and effective antidepressant treatments generally reduce immobility duration.

The sucrose preference test is used to assess anhedonia, a core symptom of depression characterized by the loss of interest or pleasure in previously rewarding activities. In this test, animals are given access to both water and a sucrose solution, and their preference for the sweet solution is measured. Reduced sucrose preference is interpreted as anhedonic behavior, and antidepressant treatments typically restore normal sucrose consumption.

2.7 Drug-Induced Depression Models

While most animal models of depression involve stress-based paradigms or genetic manipulations, pharmacologically induced depression models offer unique advantages for studying specific aspects of depressive pathophysiology. These models allow for precise control over the timing, duration, and mechanism of depression induction, making them valuable tools for investigating the neurobiological basis of mood disorders and evaluating potential therapeutic interventions.

Reserpine-induced depression represents one of the classical pharmacological models of depression. Reserpine depletes monoamine neurotransmitters (serotonin, norepinephrine, and dopamine) by irreversibly blocking the vesicular monoamine transporter, leading to behavioral and physiological changes reminiscent of human depression. This model has been instrumental in the development of the monoamine hypothesis of depression and continues to be used in preclinical research.

Chronic administration of antipsychotic medications, particularly typical antipsychotics with potent dopamine receptor antagonism, has been reported to induce depressive-like behaviors in laboratory animals. These models are particularly relevant for studying the mechanisms underlying antipsychotic-induced depression and for developing strategies to mitigate these adverse effects.

The present study builds upon this foundation by systematically investigating the depressive effects of haloperidol administration in the context of catalepsy induction, providing a unique model that captures both the motor and mood-related consequences of dopaminergic blockade.